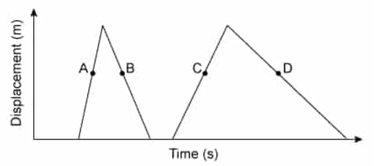
A tennis ball is thrown and bounced off a wall twice. The displacement of the tennis ball over time is shown.

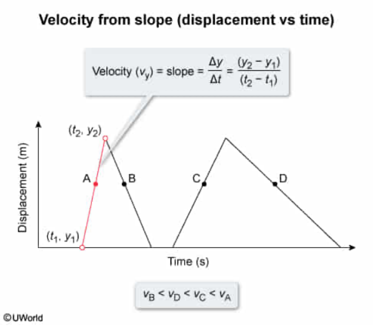


Which point on the plot corresponds to the greatest instantaneous velocity?

- A. Point A [83%]
- B. Point B [5%]
- C. Point C [3%]
- D. Point D [6%]

Correct
83% answered correctly

Explanation:



Displacement characterizes the motion of an object between two points within three-dimensional space. Unlike distance, which is the magnitude of displacement, displacement is a vector quantity with both magnitude and direction. Therefore, an object may travel some nonzero distance over the course of a physical process, but displacement will be zero if the object's initial and final position are the same.

The *x*- and *y*-components of displacement (Δx or Δy) of an object over some time interval (Δt) gives the components of **velocity (*v*)**, a vector quantity that quantifies the rate and direction of motion:

$$v_x = \frac{\Delta x}{\Delta t}$$

$$v_y = \frac{\Delta y}{\Delta t}$$

On a plot of displacement (*y*-axis) vs. time (*x*-axis), the slope of the line at each point corresponds to the **instantaneous velocity** of the object at the moment in time *t*, whereas the slope over a segment with time interval Δt corresponds to the **average velocity**.

$$v_y = \text{slope} = \frac{\Delta y}{\Delta t} = \frac{(y)_2 - y_1}{(t_2 - t_1)}$$

Accordingly, a greater (steeper), positive slope indicates a greater velocity.

In the displacement vs. time plot of this question, each line segment represents a portion of tennis ball travel. Of the four points indicated, Point A lies on the line segment with the steepest, positive slope. Therefore, the instantaneous velocity of the tennis ball is greatest at Point A, and the values of the velocities, ranked from least to greatest, are:

$$v_B < v_D < v_C < v_A$$

(Choices B and D) The slope of the line segments containing Points B and D is negative. Therefore, the instantaneous velocity of the tennis ball at these points is *less* than the velocity of the tennis ball at Points A and C.

(Choice C) The slope of the line segment containing Point A is steeper than the slope of the segment containing Point C. Therefore, the instantaneous velocity of the tennis ball at Point C is *less* than the instantaneous velocity at Point A.

Educational objective:
Displacement quantifies the movement of an object relative to a reference point. The instantaneous velocity and average velocity of an object may be determined from a plot of displacement vs time.

Time spent:0 sec
Subject:
Physics

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System:
Mechanics and Energy